

Modular Deck Drain (DD) &/or Deck Drain⁺ Evacuator (DD+E) Maintenance Manual



Modular Deck Drain (DD) and/or Deck Drain+ Evacuator (DD+E)

Refer to Project Drawing(s)

Maintenance - Deck Drain and/or Evacuator System (if applicable)

- The Deck Drain system on your pool is Modular Deck Drain including HDPE Grating.
- The Evacuator and Deck Drain system on your pool is a Modular Deck Drain with Evacuator including HDPE Grating.
- These needs to be checked and cleaned regularly.
- **The Evacuator should run 24 hours per day, 365 days per year.**

Owner's Guidelines for Evacuator System

Maintenance for Modular Deck Drain System (DD) /Deck Drain + Evacuator System

- Rinse with fresh water after cleaning the pool deck to ensure no dirt or debris is left on the surface on the grating.
- Large debris should not be washed into deck drain. The risk of poor drainage or drain blockage could create an issue of water pooling on the deck and become a hazard to pool staff and patrons.
- Using a 3/16" hex driver, check the cam-lock screws occasionally to ensure they are still tightened as desired. Cam-lock screws are stainless steel material, be careful not to over-tighten.
- If stains are present, use a cleaner such as Soft Scrub® and a nylon scrub brush. Scrub lightly as excess force could mar the finish. Do not use 100% chlorine, as commonly used to treat a pool, to clean a persistent stain.
 - Do not use MEK, turpentine, or naphtha solvent.
 - For additional information see **Grating HDPE Care & Cleaning Maintenance**.

The Evacuator should run 24 hours per day, 365 days per year.

GRATING/SHEETING - High Density Polyethylene (HDPE)

CARE AND CLEANING MAINTENANCE

Care and Maintenance

- Paddock's HDPE grating/sheeting is marine-grade polymer sheeting which is resistant to most chemicals and requires little maintenance to keep it looking new.
- Keep harsh solvents, acids, wood stains and wood preservatives away from Paddock HDPE finish. They may cause permanent staining and damage.
- Staining from rust is very difficult to remove so keep uncoated non stainless metals away from HDPE material.
- Shoes with marking soles can leave scuff marks that can be difficult to remove.

Cleaning

- For daily cleaning of everyday dirt and stains, use a non-abrasive cleaner such as "Zud" or "Soft Scrub" and a nylon brush to scrub lightly. Excessively hard scrubbing can mar the finish.
- Hard to remove stains will usually go away if you soak the area with bleach. **When using this method, you must remove the grating from the stainless steel so it will not cause corrosion. Do not** use 100% granular or tablet chlorine as a bleach alternate.
- Pressure washing can mar the finish.
- Petroleum based stains can be difficult to remove, soak the area with WD-40®. Then use acetone or toluene with a white nylon scrub pad (color-based pads can transfer dye color to finish when combined with acetone).
- **Do not** use MEK, turpentine or naphtha solvent.
- **Do not** use polishes such as Armor-All® to non-skid areas, this will reduce the slip coefficient and cause a fall/slip hazard.

Over-All Pool Maintenance

Clean facilities are safer, more attractive, and provide a more enjoyable experience for all.

Pool

- Refer to the **Safety Information** sheet.
- Vacuum regularly to keep organics off pool bottom.

Pool Deck

- Wash down pool deck regularly to remove organics that are brought into the facility by patrons & guests, which are also produced as a result of facility use by patrons.
- Use non-oxidizing cleaning solutions to clean deck.
- Clean deck drains periodically to remove build-up of organics.

Pool Deck Equipment

- Rinse and clean deck equipment on a regular basis to remove build-up of organics and corrosive condensate from chloramines and DBPs.
- Periodically wipe down stainless steel goods to remove caustic residue. Apply polish to stainless steel deck equipment, doors, water fountains, and any other fixtures that may be affected on a scheduled basis.

Patron Hygiene Program

- An effective patron hygiene program will greatly improve air and water quality, along with the staff and user experience.
- Post facility signage to encourage good personal hygiene program
- Swim Coaches can give swimmers breaks every 30 to 60 minutes. Urine is a major contributor to organic loading of a pool and a source of eye and skin irritation.
- Parents can schedule “out of pool” snack time that gives children a chance to use the restrooms.
- Facility Managers can schedule “Adult Only” swim time every hour.
- Everyone can help by encouraging showering and bathroom use before entering the pool, water park, spa, or any body of water within an aquatic facility.
- Maintain a “clean pool deck” policy: no street shoes, glass, food allowed on deck, etc.

Air Quality Recipe for Success with Evacuator System

Introduction:

Most indoor swimming environments pose a challenge for owners and operators due to the multitude of factors that are required to provide a safe, comfortable swimming environment. Treating pool water with an oxidizing sanitizer is a must for maintaining healthy pool water. Unfortunately, there is a resulting reaction which causes issues for most indoor swimming environments.

All pools generate off-gassing Disinfectant By-Products (DBPs) as a result of oxidizing chemicals (chlorine/bromine) doing what they are supposed to do: *Oxidize Organics*. Organics could be defined as common foreign objects introduced to the pool: dirt, hair, skin cells, saliva, urine, etc. The issue relative to indoor pools and oxidizing organics is where these DBPs go and what impact they are having on the following: Patrons & Staff, Facility & Replacement Costs, Operating Costs, and Reputation.

Facts:

- Swimmers introduce organics into pools and onto pool decks which must be oxidized in order to maintain a healthy environment.
- The oxidation process results in DBPs off-gassing from the chlorinated pool water, including water splashed onto pool decks as well as aerosolize through water features.
- Traditional indoor aquatic facility HVAC systems are designed to heat, cool, and/or dehumidify natatorium air. They also are designed to introduce outdoor air (dilution) on an as-needed basis.
- The combination of chlorine oxidizing organics, DBPs off-gassing from the water's surface & pool deck, and the HVAC re-circulating DBP-laden air, creates a compounding effect of the contaminated air, which all contribute to a "Poor Indoor Air Quality Cycle".

Water & Air: Bridging the Gap

A swimming pool's water purification system, operating procedures, and resulting water quality conditions serve as the foundation for a natatorium's overall environment. Water temperature, filtration effectiveness, water chemistry conditions, chemical feed systems, types of chemicals, supplemental disinfecting equipment, facility maintenance procedures, and the organic load in pool water (clarity is an indicator) all contribute to the health of the pool water in both a positive and/or negative way. A swimming pool is a living biological 'science project' and requires consistent professional maintenance to stay ahead of compounding issues.

A natatorium's mechanical system provides patron comfort in the natatorium. This depends on operating parameters, equipment condition, and engineered airflow. The equipment must be designed properly and maintained on a regular basis to ensure proper effective operation.

Facility Air Considerations

Mechanical System Design & Operation

- Proper Airflow Rates:
 - Total Airflow
 - Supply Air
 - Exhaust Air
 - Outdoor Air
 - Evacuator Exhaust
- Duct System Design
 - Proper Return & Supply Distribution
 - Avoid Dead Spots, Stratification, & Zonal Effects
 - May Need Supply Register Adjustments
 - May Need Supplemental Directional Fans

Design Parameters

Secondary Facility Air Disinfection

- Cold Plasma Ionization: Located within the Pool Dehumidification Unit (PDU), this system disinfects and reduces VOC (volatile organic compound) levels in the air being recirculated in the indoor aquatic environment. This technology creates positive and negative ions, which are drawn to airborne particles due to their polarization, making them larger and increasing the effectiveness of filtration. In the same manner, they attach to pathogens such as viruses, bacteria, and mold spores, robbing and deactivating the hydrogen component of the pathogen creating a cleaner/safer environment. It also has additional air quality improvements such as odor control and reducing allergens. The use of the Cold Plasma system will also allow you to run your outside air at lower levels when conditions warrant it, saving energy.

System Balance:

Paddock Evacuator Technology is designed to complement the operation of the existing mechanical system. Upon installation of Evacuator Technology, the owner/operator should consult his servicing mechanical contractor to balance the existing mechanical system's exhaust rate with that of the Evacuator.

Balance Mechanical System to:

- Chloramines are produced and released into the air in aquatic facilities at all times, even during unoccupied times.
- A recommended 0.05"—0.15" WC negative pressure must be maintained in the natatorium air space at all times. This keeps the 'pool [corrosive air] in the pool [environment].'
- Maintain outdoor air introduction through the existing mechanical system as designed by a professional mechanical engineer and/or manufacturer's recommendation and at least the ASHRAE minimum outdoor air standard, unless otherwise specified.
- **Opening doors and windows are not recommended, as this can overload the mechanical equipment, interrupt the intended air flow patterns, and/or create an environment with bad (or worse) air conditions.**
- Owner/operator may reduce exhaust air further in the existing mechanical system during off-hour / unoccupied times to recover energy costs as design allows. Technology now exists that allows facilities to run below ASHRAE minimum outdoor air volumes based on certain criteria.
- Reduce the exhaust rate of the existing mechanical system as required. In some cases, the exhaust fan(s) in the existing mechanical system may be shut down completely as Evacuator Technology may handle 100% of the required exhaust.
- Source-capture exhaust removes a much higher amount (percentage) of airborne chloramines and other disinfection by-products per CFM exhausted than conventional return exhaust.
- Turn off / disconnect supplementary exhaust fans in the space, per design. The majority of exhaust will typically be through the Evacuator, unless otherwise designed. Please check with your installer or the supplied drawings to verify this information.
- **The Evacuator should run 24 hours per day, 365 days per year.**

Owners Responsibility & Suggested Operating Guidelines to Complement Evacuator Technology:

- Ensure Evacuator is running at all times: 24 hours per day, 365 days per year. If Evacuator is deemed to not be operational, contact Paddock Pool Equipment Co. or your local mechanical representative immediately.
- Reduce organics introduced into the pool area by enforcing aggressive patron hygiene and maintenance programs.
- Maintain pool water chemistry based on the guidelines provided above.
- Maintain Evacuator, Pool, & Mechanical System as outlined above for best results.
- Balance existing mechanical system to ensure a recommended 0.05"—0.15" WC negative pressure exists in the facility.



- Maintain existing mechanical system to ensure proper airflow and outdoor air introduction as designed per the facilities operating parameters.
- HVAC Maintenance Company / Mechanical Contractor is scheduled to provide routine maintenance as defined by the mechanical equipment manufacturer. Preventative maintenance is the key!